Deep Learning Based Medical Image Analysis For Early Disease Detection

Technical Documentation

Phase III project submission | 30 October 2026

Student / institution: enter the approved cover details. No identity or approval has been invented.

Abstract, problem statement and objectives

The approved Phase-I report investigates early disease detection support for thoracic findings in chest radiographs, diabetic-retinopathy severity in retinal fundus images, and skin lesions in dermoscopic images. This implementation provides separate deep-learning image classification pipelines and a local interactive research app. It is not a diagnostic product.

The report is Phase I rather than an exact final experimental protocol. It does not determine releases, exact model selection, training hyperparameters, split proportions or performance targets. These choices are recorded as implementation decisions rather than attributed to the report.

Scope and implementation checklist

Requirement traceability is maintained in IMPLEMENTATION_CHECKLIST.md. ISIC 2016 Task 3 was selected because the report says ISIC without identifying a release. NIH and APTOS remain coded but await actual local dataset access. U-Net is not used because no segmentation masks or segmentation target are specified.

Requirements and technology

Category	Implementation
Runtime	Python 3.11, virtual environment
ML	PyTorch, torchvision, scikit-learn
UI	Streamlit
Data/visualization	Pillow, NumPy, Pandas, Matplotlib
Hardware verified	Windows; NVIDIA RTX 3050 Laptop GPU, CUDA available; 6 GB VRAM
Storage / deployment	Datasets and checkpoint remain local; deployment URL pending

System architecture and data flow

Dataset metadata adapter -> validated image records -> deterministic train/validation/test partition -> common image transform -> transfer-learning classifier -> training-only class weights and validation-loss checkpoint selection -> held-out evaluation -> serialized checkpoint and metrics -> Streamlit prediction and optional Grad-CAM.

ISIC official test records remain fixed; validation is split from source training records. No patient identifiers are present in the selected task label files. The independent test set is not used for selection.

Algorithms and evaluation

ResNet-18, ResNet-50, and DenseNet-121 are configurable transfer-learning classifiers. The completed experiment uses pretrained ResNet-18 and binary softmax cross-entropy. Metrics include accuracy, macro precision/recall/F1, one-vs-rest ROC-AUC, per-class sensitivity/specificity and confusion matrix. Training set class weights address the observed imbalance. Seed: recorded in experiment metadata.

Module descriptions

`src/data/` loads task metadata and labels; `src/datasets.py` exposes the PyTorch dataset; `src/preprocessing.py` validates inputs and defines transforms; `src/models.py` builds classifier backbones; `src/training.py` trains and checkpoints; `src/evaluation.py` computes metrics; `src/prediction.py` performs inference; `src/explainability.py` creates Grad-CAM; `src/pipeline.py` orchestrates the experiment; `app.py` is the Streamlit UI; `train.py` and `audit_data.py` are command-line entry points.

Security, error handling and limitations

Raw image data and checkpoint files are excluded from ordinary Git tracking. The app enforces upload size/type/readability checks and handles missing model artifacts. No API keys are required. Medical images should remain de-identified. Dataset shift, label quality, class imbalance and unknown patient separation limit generalization. Outputs are experimental and not clinically validated.

Future work and conclusion

Obtain the complete NIH release after provisioning adequate storage; accept APTOS rules and download via the user's Kaggle account; audit both datasets; train using approved splits; add comparative backbone runs and repeat-seed uncertainty analysis; review dataset-specific licensing; deploy only with suitable privacy controls and hosting resources.

References

Supplied Phase-I project report, `report.docx` (local source specification; intentionally not published because it contains academic identity details).

ISIC Challenge data: https://challenge.isic-archive.com/data/

NIH Clinical Center ChestX-ray dataset: https://nihcc.app.box.com/v/ChestXray-NIHCC

APTOS 2019 competition data and rules: https://www.kaggle.com/competitions/aptos2019-blindness-detection/data

PyTorch installation selector: https://pytorch.org/get-started/locally/